

# Announcements – Feb 9

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- HW 4 is due Friday at 9am. HW 4 is only odd numbered problems. It is given to help you prepare for the exam.
- Today's "quiz" will be like the last one. You will need a partner.
- SI sessions are Tue 6-7pm and Thu 2-3pm.
- Exam 1 is Friday next week. It will cover chapter 1. Wednesday of next week our class session will be a review. My exam questions are all very similar to homework questions, so the best way to study is to look at the homework.
- Homework solutions are being posted on D2L. They are the work of one of our dedicated TAs! 😊

# Uniqueness Proofs

- To prove that there is a unique element that makes something true:  
 $\exists! xP(x)$   
you must prove two things.
  - 1. There is an  $x$  that makes  $P(x)$  true. (Existence)
  - 2. If  $P(y)$  is true for any element, then  $y = x$ . (Uniqueness)

## Uniqueness Proof Example

Prove that for every integer  $a$ , there is exactly one integer  $b$  such that  $a + b = 0$ .

Pf: Let  $a$  be an integer.

Then  $-a$  is an integer ( $-0 = 0$ ) and

$$a + (-a) = 0 \quad (\text{existence proved})_/$$

Suppose  $b$  is an integer such that  $a + b = 0$

$$\Rightarrow a + b - a = 0 - a$$

$$\Rightarrow b = -a \quad (\text{uniqueness proved})_{//}$$

- Suppose we have a conjecture that we want to either prove or disprove.
- First, try some examples. If you find a counterexample, then you have disproved the conjecture and you're done.
- If the examples work, then perhaps the conjecture is true. So how do you find a proof?
- Are the number of possibilities finite and small? If so, an exhaustive proof might be the way to go.
- Does it make sense to break the problem into cases? If so, each of those cases might be proved different ways.

- Look at doing a direct proof.
- If you have trouble getting a direct proof, look at doing a proof by contraposition or contradiction.
- What if you are still stuck?
- One thing you might try is backward reasoning.
- If you're trying to prove  $p \rightarrow q$ , you might get some insight by starting with  $q$  and working backward.
- What you want to find is some proposition  $r$ , such that you can prove  $p \rightarrow r$  and  $r \rightarrow q$

## Backward Thinking Example

For real numbers  $x$  and  $y$ , the arithmetic mean (average) is defined as  $\frac{x+y}{2}$  and the geometric mean is defined as  $\sqrt{xy}$

Prove that for all positive, distinct real numbers, the arithmetic mean is greater than the geometric mean.

Let's try some examples  $(8+2)/2 = 5$ ,  $\sqrt{8 \times 2} = \sqrt{16} = 4$  and  $5 > 4$

Do some more on your own.

## Backward Thinking Example

Prove that for all positive, distinct real numbers, the arithmetic mean is greater than the geometric mean.

Let's do some backward thinking. We'll start with what we want to prove.

$$\frac{x+y}{2} > \sqrt{xy}$$

$$\frac{(x+y)^2}{4} > xy \quad \text{-square both sides}$$

$$(x+y)^2 > 4xy \quad \text{-multiply both sides by 4}$$

$$x^2 + 2xy + y^2 > 4xy \quad \text{-arithmetic}$$

$$x^2 - 2xy + y^2 > 0 \quad \text{-subtract } 4xy \text{ from both sides}$$

$$(x-y)^2 > 0 \quad \text{-factor}$$

The last is true as long as  $x$  and  $y$  are distinct. So, the question is, are the steps reversible?

Yes, except to go from the 2<sup>nd</sup> to the 1<sup>st</sup> we need to know both sides are not negative.

We can now write out the proof as:

## Backward Thinking Example

Prove that for all positive, distinct real numbers, the arithmetic mean is greater than the geometric mean.

Pf: Let  $x$  and  $y$  be distinct, positive real numbers

$$\Rightarrow (x - y)^2 > 0 \quad \text{-since } x \text{ and } y \text{ are distinct and squares non-negative}$$

$$\Rightarrow x^2 - 2xy + y^2 > 0 \quad \text{-arithmetic}$$

$$\Rightarrow x^2 + 2xy + y^2 > 4xy \quad \text{-add } 4xy \text{ to both sides}$$

$$\Rightarrow (x + y)^2 > 4xy \quad \text{-factor left side}$$

$$\Rightarrow \frac{(x+y)^2}{4} > xy \quad \text{-divide both sides by 4}$$

$$\Rightarrow \frac{x+y}{2} > \sqrt{xy} \quad \text{-take the square root of both sides}_{//}$$



## Backward Thinking WARNING

- Backward thinking can be very dangerous, particularly if you are new to proofs.
- Remember  $p \rightarrow q$  is NOT the same as  $q \rightarrow p$  !!!
- The proof in the example worked because almost all of the steps were double implications. That will not generally be true.
- Backward thinking is very useful if you're trying to show two expressions are equal.
- It can give insight in other cases, but please use it with caution.

- Another strategy for trying to figure out a proof, is to look at existing proofs of existing results.
- While you can't use one of these directly, you might be able to make simple alterations to come up with a proof for your current problem.
- Even if not, you might get insight into how to prove the result you want.

Say you're trying to prove that  $\sqrt{3}$  is irrational.

We have the proof that  $\sqrt{2}$  is irrational:

By contradiction:

Assume  $\sqrt{2}$  is rational

$\sqrt{2} = \frac{a}{b}$  where  $a$  and  $b$  are integers with no common factors,  $b \neq 0$

$$\Rightarrow 2 = \frac{a^2}{b^2}$$

$$\Rightarrow 2b^2 = a^2$$

$$\Rightarrow a^2 \text{ is even} \Rightarrow a \text{ is even}$$

$$\Rightarrow a = 2k, k \in \mathbb{Z}$$

$$\Rightarrow 2b^2 = (2k)^2 = 4k^2$$

$$\Rightarrow b^2 = 2k^2$$

$$\Rightarrow b^2 = 2k^2$$

$$\Rightarrow b^2 \text{ is even} \Rightarrow b \text{ is even}$$

$\Rightarrow$  both  $a$  and  $b$  are even, so they have a common factor of 2

$$\Rightarrow \Leftarrow$$

$\therefore \sqrt{2}$  is irrational

How can we adapt this for  $\sqrt{3}$  ?

## Proof Strategies – Similar Proof Example

Prove that  $\sqrt{3}$  is irrational:

By contradiction:

Assume  $\sqrt{3}$  is rational

$\sqrt{3} = \frac{a}{b}$  where  $a$  and  $b$  are integers with no common factors,  $b \neq 0$

$$\Rightarrow 3 = \frac{a^2}{b^2}$$

$$\Rightarrow 3b^2 = a^2$$

$\Rightarrow 3$  divides  $a^2$  -since  $b^2 = \frac{a^2}{3}$  and  $b^2$  is an integer

$$\Rightarrow a^2 = 3j, j \in \mathbb{Z}$$

$$\Rightarrow \frac{a}{3} \cdot \frac{a}{1} = j$$

$\Rightarrow 3$  divides  $a$  - since  $j$  is an integer and 3 cannot be factored.

$$\Rightarrow a = 3k, k \in \mathbb{Z}$$

## Proof Strategies – Similar Proof Example

$$\Rightarrow 3b^2 = (3k)^2 = 9k^2$$

$$\Rightarrow b^2 = 3k^2$$

$$\Rightarrow b^2 = 3k^2$$

$$\Rightarrow 3 \text{ divides } b^2 \Rightarrow 3 \text{ divides } b \text{ (same reasons as for } a \text{ above)}$$

$$\Rightarrow 3 \text{ divides both } b \text{ and } a$$

$$\Rightarrow \Leftarrow$$

$$\therefore \sqrt{3} \text{ is irrational}$$